Composite Disturbance Filtering for Interaction Force Estimation With Online Environmental Stiffness Exploration

Yanran Wei, Jiayin Wang, Wenshuo Li, Xiaofan Du, Xiang Yu, *Senior Member, IEEE*, and Lei Guo, *Fellow, IEEE*

***Abstract*—The primary focus of this article is the interaction force estimation of robotic manipulators with environmental stiffness identification and exploration. This issue is particularly crucial in minimally invasive surgery, where the interaction force between the end effector and the soft tissues needs to be estimated. Existing methods primarily focus on observer design, exploiting only the information from the robot dynamics. To utilize more information, a unified force estimation framework is proposed in this article, where the robot dynamics and force generation model are simultaneously taken into account. Specifically, a robot–environment coupled system model is established by regarding the interaction force as a state-coupled disturbance of the robot system. Based on this, a separability analysis for the interaction force is conducted. To cope with the unknown stiffness parameter and stochastic uncertainties, a novel composite disturbance filtering scheme is developed. An expectation-maximization-based environmental stiffness exploration force observer (EEFO) is constructed for simultaneous environmental stiffness identification and interaction force estimation. The performance of the proposed EEFO is evaluated via numerical simulations and experimental tests on a surgical robot platform. The results have demonstrated the superiority of the proposed scheme over the state-of-the-art methods.**

***Index Terms*—Composite disturbance filtering (CDF), interaction force estimation, medical robots and systems, robot–environment interaction, unknown deformable environment.**

Manuscript received 1 February 2024; revised 9 May 2024 and 13 July 2024; accepted 5 August 2024. Date of publication 30 August 2024; date of current version 18 June 2025. Recommended by Technical Editor A.-T. Nguyen and Senior Editor H. Fujimoto. This work was supported in part by the National Natural Science Foundation of China under Grant 62373033, Grant 62388101, Grant 62122007, and Grant 62103013, in part by the Natural Science Foundation of Zhejiang Province under Grant LZ23F030011 and Grant LD21F030001, in part by the Defense Industrial Technology Development Program of China under Grant JCKY2020601C016, and in part by the Key Research and Development Program of Zhejiang Province under Grant 2021C03158. *(Corresponding author: Wenshuo Li.)*

Yanran Wei, Xiaofan Du, Xiang Yu, and Lei Guo are with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China (e-mail: yrweibuaa@126.com; xiaofandu@buaa.edu.cn; xiangyu_buaa@buaa.edu.cn; lguo@buaa.edu.cn).

Jiayin Wang is with the School of Electronic Information Engineering, Tongji University, Shanghai 200092, China, and also with the MicroPort MedBot (Group) Company Ltd., Shanghai 200137, China (e-mail: wangjy@microport.com).

Wenshuo Li is with the Hangzhou Innovation Institute, Beihang University, Hangzhou 310051, China (e-mail: wslibuaa@126.com).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TMECH.2024.3443310.

Digital Object Identifier 10.1109/TMECH.2024.3443310

I. Introduction

THE field of robotics is rapidly evolving, with robots increasingly engaged in complex manipulation tasks that involve direct interaction with the external environment. Typical scenarios include minimally invasive surgery [1] and in-orbit assembly of space manipulators [2]. High-precision acquisition of the interaction force is crucial to ensure safety and efficiency of these tasks. The problem of acquiring interaction forces has gathered significant research attention due to its practical importance. To avoid the additional costs and workload of using force sensors, model-based force estimation methods are often preferred. These methods treat the interaction force as an unknown input or disturbance in the dynamic model of the robot manipulator.

Many elegant results on force observer design have been achieved, including frequency-domain Disturbance observers (DO) [3], [4], [5], time-domain nonlinear DOs (NDO) [6], [7], [8], generalized momentum observers (GMO) [9], [10], filter-based methods [11], [12], [13], [14], [15], and neural network (NN)-based observers [10], [16], among others. The force signal is often submerged within other disturbances such as model uncertainties (e.g., friction [17], actuator saturation [18]) and stochastic noises. The key to model-based disturbance estimation is the refined quantification and separation of these composite disturbances [19], [20]. An adaptive DO in [19] separates disturbance torque from nominal torque during target capture task. In [12], a force estimation approach uses a dither periodic component elimination Kalman filter (KF) to quantify and separate stochastic noises and static friction. A detailed experimental comparison of model-based and NN-based methods for force estimation is provided in [21]. In [22], a semiparametric model-based disturbance KF (DKF) facilitates the separation of model uncertainties. Our recent work [16] enhances this approach by learning coupled multiple disturbances, quantifying modeling uncertainties with an enhanced Gaussian Process (GP) model, and developing an adaptive DKF within the composite disturbance filtering (CDF) framework [23] for refined separating interaction force from composite disturbances.

It should be noted that the aforementioned methods only exploit information from the robot dynamics. In many practical

1083-4435 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

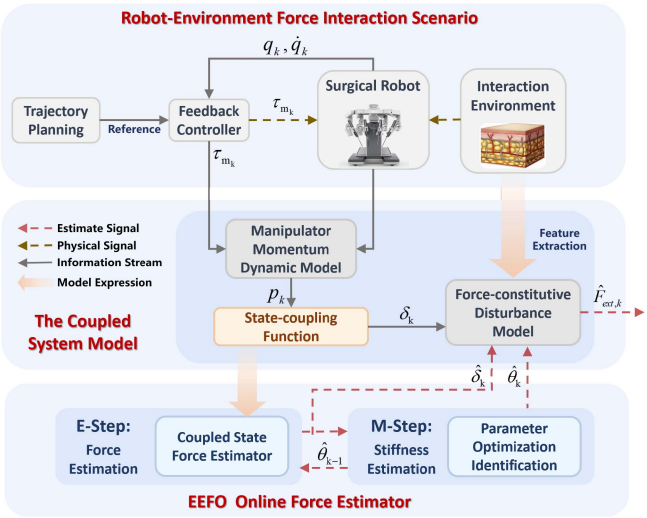


Fig. 1. Illustration of the proposed EEFO force estimation scheme.



Fig. 2. Surgical robot platform interacting with a silicone kidney model. Toumai laparoscopic surgical robot (MicroPortToumai, CN) [31].

scenarios, environmental characteristics can also be utilized. For instance, in surgical robots interacting with soft tissue [24], more accurate interaction force estimates can be achieved by incorporating environmental stiffness information from force-constitutive models. These models characterize the dynamic relationship between environmental surface deformation and interaction force, forming the foundation of the Environmental characteristics Exploration (EE) force estimation scheme. Several methods using additional force sensors for force estimation based on the EE scheme have been proposed [24], [25], [26], [27], [28]. For scenarios without force sensors, stiffness estimation approaches have been presented in [13] and [29], where environmental stiffness is treated as an augmented state of the robot system and estimated by Extended KF (EKF). However, these methods still depend on direct position measurements of the end-effector and the controller, which can introduce errors in force estimation due to controller errors. As a result, the estimation error will accumulate or even diverge [30].

The challenge of simultaneously estimating interaction forces and identifying stiffness parameters remains largely unexplored. Existing observer-based force estimation methods mainly treat the interaction force as an unknown input in the robot dynamics, neglecting environmental characteristics such as stiffness, which also carry force information. Conversely, environmental stiffness estimation approaches rely on additional force or position measurements on end effectors, which is impractical in certain applications. To address these limitations and simultaneously exploit information from robot dynamics and environmental stiffness, a challenging problem is the simultaneous estimation of interaction force and stiffness parameters. To this end, an expectation-maximization (EM)-Bayesian filter-based *environmental stiffness exploration force observer (EEFO)* method is proposed in this article. The overall architecture of the proposed scheme is illustrated in Fig. 1. The middle area details the robot–environment coupled system model established in this work, underpinning the force estimation. This model incorporated the dynamics of the robot manipulator, the force-constitutive model with environmental stiffness, and their coupled relationship. The lower area describes the EEFO online force estimator, which iterates between the E-Step and the M-Step to update the state $\hat{\delta}_k$ of the coupled system and the stiffness parameter $\hat{\theta}_k$, ultimately producing force estimate $\hat{F}_{d_k}$. The superiority and practicality of the proposed scheme are validated through numerical simulations and experimental tests on a Toumai surgical robot platform (MicroPortToumai, CN) (see Fig. 2).

The contribution of this article can be summarized as follows.

1) By simultaneously exploiting the information from robot dynamics and force generation models, a unified framework for the interaction force estimation is proposed. A robot–environment coupled system model that integrates both the robot dynamics and environmental stiffness is established, which differs from the existing model [7], [9] where only the robot dynamics are considered. Furthermore, the separability conditions of the interaction force signal are established. On this basis, a reliable force estimate can be obtained by constructing an interaction force observer with real-time environmental stiffness exploration.
2) To address the unknown environmental stiffness and stochastic uncertainties in force estimation, a novel EEFO method is developed based on the robot–environment coupled system model. In the proposed EEFO, the interaction force signal and the unknown stiffness parameter can be jointly estimated via an iterated EM procedure. Different from the existing stiffness identification schemes [24], [26], where direct force measurement is required, the proposed method has treated the interaction force as a hidden state which can be estimated along with the stiffness parameter with guaranteed noise attenuation performance. By this means, the requirement for additional force sensing devices can be circumvented.
3) The proposed framework has been evaluated via numerical simulations, and experimentally validated on a surgical robot with tissue-like soft materials. Both simulation and experimental results have shown that the proposed scheme outperforms the state-of-the-art force estimation methods including the DO [3], NDO [7], GMO [9],

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

KF [11], and DKF [22] in terms of estimation accuracy and noise attenuation degree. Moreover, direct force and position measurements are no longer required, which broadens the applicability scopes of the proposed scheme.

The rest of this article is organized as follows. In Section II, the robot–environment coupled dynamic system model is established, and the force separability analysis is also proposed. In Section III, the EEFO method is elaborated and its superiority over the existing schemes is demonstrated via Monte Carlo simulation. Experimental results obtained in the robot–silicone interaction scenario are presented in Section IV. Finally, Section V concludes this article.

II. Robot–Environment Coupled System: Modeling and Separability Analysis

In this section, a robot–environment coupled system model is established, where the information from the robot dynamics and force-constitutive model is simultaneously exploited. In Section II-A, the manipulator dynamics are established in joint generalized momentum space. As detailed in Section II-B, an interaction force generation model with unknown stiffness parameters is formulated to characterize the explicit functional relationship between the interaction force and the generalized momentum of the robot manipulator. In Section II-C, the robot dynamics and the force generation model are integrated, yielding the robot–environment coupled system model. This coupled model provides the foundation for the derivation of EEFO in Section III. Furthermore, separability analysis is presented.

A. Manipulator Dynamics Model

The dynamic model of an n-link serial manipulator can be expressed as [32] follows:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{G}(\boldsymbol{q}) + \boldsymbol{\tau}_f(\boldsymbol{q},\dot{\boldsymbol{q}}) = \boldsymbol{\tau}_m + \boldsymbol{\tau}_{\text{ext}}, \quad (1)$$

where $\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}} \in \mathbb{R}^n$ denote the joint position, velocity, and acceleration, respectively; $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n\times n}$ is the symmetric positive-definite inertia matrix; $\mathbf{C}(\mathbf{q},\dot{\mathbf{q}})\dot{\mathbf{q}} \in \mathbb{R}^n$ represents Coriolis and centripetal effects; $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^n$ is the gravity vector; $\boldsymbol{\tau}_f \in \mathbb{R}^n$ denotes the joint torque due to inner friction; $\boldsymbol{\tau}_m \in \mathbb{R}^n$ denotes the active motor torque; $\boldsymbol{\tau}_{\text{ext}} \in \mathbb{R}^n$ represents the equivalent external torque from the generalized interaction force $\boldsymbol{F}_{\text{ext}} \in \mathbb{R}^{n_d}$. $\boldsymbol{F}_{\text{ext}}$ is the wrench exerted on the tool center point (TCP) of the end effector that can be defined in the Cartesian coordinates as

$$\boldsymbol{F}_{ext} = [f_x, f_y, f_z, \tau_x, \tau_y, \tau_z]^T, \quad (2)$$

where $f_x, f_y, f_z \in \mathbb{R}$ are the interaction forces, and τ_x, τ_y, τ_z are the interaction torques. The relationship between joint torque vector and wrench vector is

$$\boldsymbol{\tau}_{\text{ext}} = \boldsymbol{J}(\boldsymbol{q})^T \boldsymbol{F}_{\text{ext}}, \quad (3)$$

where $\boldsymbol{J}^T(\boldsymbol{q}) \in \mathbb{R}^{n\times n_d}$ is the transpose of the Jacobian matrix.

Defining the generalized momentum of the robot manipulator as $\boldsymbol{p} := \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}}$, it follows that:

$$\dot{\boldsymbol{p}} = \dot{\boldsymbol{M}}(\boldsymbol{q})\dot{\boldsymbol{q}} + \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}}, \quad (4)$$

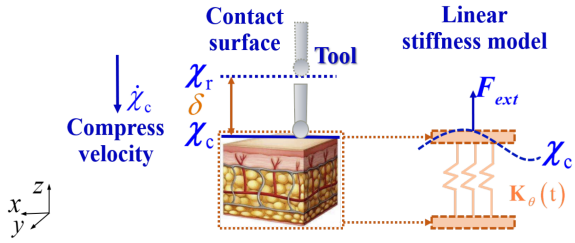


Fig. 3. Graphical illustration of the interaction model between deformable environment and end-effector tool.

where joint acceleration $\ddot{\boldsymbol{q}}$ is involved. In order to circumvent the need for direct measurement of $\ddot{\boldsymbol{q}}$, which may introduce additional noises, we utilize the skew-symmetric property of $\dot{\boldsymbol{M}}(\boldsymbol{q}) - 2\boldsymbol{C}$ and write it into an equivalent form [9], [32]

$$\dot{\boldsymbol{M}}(\boldsymbol{q}) = \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{C}^T(\boldsymbol{q},\dot{\boldsymbol{q}}). \quad (5)$$

Substituting (1) and (5) into (4), we obtain the generalized momentum dynamic equation

$$\dot{\boldsymbol{p}} = \boldsymbol{\tau}_m + \boldsymbol{\tau}_{\text{ext}} - \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})^T\dot{\boldsymbol{q}} - \boldsymbol{G}(\boldsymbol{q}) - \boldsymbol{\tau}_f. \quad (6)$$

The dynamic model (6) can be discretized as

$$\begin{cases} \boldsymbol{p}_{k+1} = \boldsymbol{A}_k\boldsymbol{p}_k + \boldsymbol{B}_k\boldsymbol{u}_k + \boldsymbol{E}_k\boldsymbol{F}_{ext,k} + \boldsymbol{w}_k, \\ \boldsymbol{y}_k = \boldsymbol{C}_k\boldsymbol{p}_k + \boldsymbol{v}_k, \end{cases} \quad (7)$$

where subscript $\cdot_k$ denotes discrete time step t_k. $\boldsymbol{w}_k$ and $\boldsymbol{v}_k$ are the process and measurement noises, respectively, satisfying $\boldsymbol{w}_k \sim \mathcal{N}(0, \Sigma_{w_k})$ and $\boldsymbol{v}_k \sim \mathcal{N}(0, \Sigma_{v_k})$. $\boldsymbol{y}_k \in \mathbb{R}^n$ is the measurement signal, and

$$\boldsymbol{u}_k = \boldsymbol{\tau}_{m,k} + \boldsymbol{C}(\boldsymbol{q}_k,\dot{\boldsymbol{q}}_k)^T\dot{\boldsymbol{q}}_k - \boldsymbol{G}_k(\boldsymbol{q}_k) - \boldsymbol{\tau}_{f,k} \quad (8)$$

is the known lumped input term. The coefficient matrices are expressed as

$$\boldsymbol{A}_k = \boldsymbol{I}_n, \boldsymbol{B}_k = \Delta T \cdot \boldsymbol{I}_n, \boldsymbol{C}_k = \boldsymbol{I}_n, \boldsymbol{E}_k = \boldsymbol{J}_k^T\Delta T.$$

B. Force Generation Model

Under the assumption of linear stiffness, isotropy of the environment and a single-point contact scenario (see Fig. 3), the constitutive model of the interaction force is given by [24]

$$\boldsymbol{F}_{ext} = \boldsymbol{K}_\theta \cdot \boldsymbol{\delta} \quad (9)$$

where $\boldsymbol{\delta} \in \mathbb{R}^{n_d}$ represents the deformation degree vector, defined as $\boldsymbol{\delta} = \boldsymbol{\chi}_c - \boldsymbol{\chi}_r$. Here, $\boldsymbol{\chi}_c$ denotes the TCP position of the end effector and the contact surface, while $\boldsymbol{\chi}_r$ denotes the position of the reference surface. The deformation degree vector depends on both the controlled movement of the robot manipulator and the autonomous movement of the surface of the dynamic object (e.g., cardiac tissue motion). Considering only interactions with static objects, it follows that $\dot{\boldsymbol{\chi}}_r = \boldsymbol{0}$. Supposing that the TCP does not penetrate the surface of the object, $\boldsymbol{\chi}_c$ can be derived from the forward kinematics of the robot manipulator as

$$\dot{\boldsymbol{\chi}}_c = \boldsymbol{J}(\boldsymbol{q}) \cdot \dot{\boldsymbol{q}} \quad (10)$$

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

which further leads to

$$\dot{\boldsymbol{\delta}} = \boldsymbol{J}(\boldsymbol{q})\boldsymbol{M}(\boldsymbol{q})^{-1}\boldsymbol{p}. \tag{11}$$

$\boldsymbol{K}_\theta \in \mathbb{R}^{n_d \times n_d}$ in (9) is a time-invariant environmental stiffness matrix. This matrix typically characterizes the material properties and geometry of the object. Under the isotropy assumption, $\boldsymbol{K}_\theta$ can be represented as a diagonal matrix that $\boldsymbol{K}_\theta = \text{diag}\{\boldsymbol{\theta}\}$, where $\boldsymbol{\theta} = [\theta_{n_1}, \ldots, \theta_{n_d}]$. This work focuses solely on interaction force estimation during the postcontact phase.

Remark 1: In this work, the constitutive model of the interaction force (9) is employed under the assumptions of linear stiffness and an isotropy environment. Although these assumptions may introduce bias between the practical force signal and the model, they significantly simplify the model. Furthermore, the fundamental idea of the EEFO scheme—iterative force estimation and stiffness (or *generalized stiffness*) identification—can also be applied to more complex constitutive models, such as the Hunt–Crossley model, Maxwell model, Kelvin–Voigt model, and Kelvin–Boltzmann model [33], [34].

Defining $\boldsymbol{\omega}_k := \boldsymbol{\delta}_k$ as the state variable of the interaction force generation system. Combining (9) and (11), and using the Euler method for discretization, the generation model of the interaction force is

$$\begin{aligned} \boldsymbol{\omega}_{k+1} &= \boldsymbol{W}_k\boldsymbol{\omega}_k + \boldsymbol{G}_k\boldsymbol{p}_k + \boldsymbol{\xi}_k \\ \boldsymbol{F}_{\text{ext},k} &= \boldsymbol{K}_\theta\boldsymbol{\omega}_k + \boldsymbol{\nu}_k. \end{aligned} \tag{12}$$

The coefficient matrices are expressed as

$$\boldsymbol{W}_k = \boldsymbol{I}_{n_d}, \quad \boldsymbol{G}_k = \boldsymbol{J}_k\boldsymbol{M}_k^{-1}\Delta T \tag{13}$$

where $\boldsymbol{\omega}_k \in \mathbb{R}^{n_d}$, and the noise terms $\boldsymbol{\xi}_k$ and $\boldsymbol{\nu}_k$ satisfy $\boldsymbol{\xi}_k \sim \mathcal{N}(0, \Sigma_{\xi_k})$ and $\boldsymbol{\nu}_k \sim \mathcal{N}(0, \Sigma_{\nu_k})$, respectively. These noise terms account for deviations between the actual and theoretical models of $\boldsymbol{F}_{\text{ext},k}$, arising from nonideal materials characteristics.

Remark 2: The relationship between the deformation degree $\boldsymbol{\delta}$ and the joint momentum of the robot has been established in (12). By this means, $\boldsymbol{\delta}$ can be obtained based on the robot kinematics, circumventing the requirement for direct position measurement. This approach is particularly suitable for applying to surgical robot, where additional position sensors are unavailable for the end-effector in contact with soft tissues within a limited operational space.

C. Robot–Environment Coupled System Model

1) Modeling the Coupled System: Letting $\boldsymbol{x}_k := [\boldsymbol{p}_k^T, \boldsymbol{\omega}_k^T]^T$, the augmented system can be modeled as

$$\begin{cases} \boldsymbol{x}_{k+1} = \bar{\boldsymbol{A}}_{\theta_k}\boldsymbol{x}_k + \bar{\boldsymbol{B}}_k\boldsymbol{u}_k + \bar{\boldsymbol{w}}_k \\ \boldsymbol{y}_k = \bar{\boldsymbol{C}}_k\boldsymbol{x}_k + \bar{\boldsymbol{v}}_k \end{cases} \tag{14}$$

where the coefficient matrices are

$$\bar{\boldsymbol{A}}_{\theta_k} = \begin{bmatrix} \boldsymbol{A}_k & \boldsymbol{E}_k\boldsymbol{K}_\theta \\ \boldsymbol{G}_k & \boldsymbol{W}_k \end{bmatrix}, \bar{\boldsymbol{B}}_k = \begin{bmatrix} \boldsymbol{B}_k \\ \boldsymbol{0} \end{bmatrix}, \bar{\boldsymbol{C}}_k = [\boldsymbol{C}_k, \boldsymbol{0}]. \tag{15}$$

The noise terms are given by

$$\bar{\boldsymbol{w}}_k = \begin{bmatrix} \boldsymbol{w}_k + \boldsymbol{E}_k\boldsymbol{\nu}_k \\ \boldsymbol{\xi}_k \end{bmatrix}, \quad \bar{\boldsymbol{v}}_k = \boldsymbol{v}_k, \tag{16}$$

which satisfy $\bar{\boldsymbol{w}}_k \sim \mathcal{N}(0, \Sigma_{\bar{w}_k})$ and $\bar{\boldsymbol{v}}_k \sim \mathcal{N}(0, \Sigma_{\bar{v}_k})$, respectively, where $\Sigma_{\bar{w}_k} = \text{diag}\{\Sigma_{w_k} + \boldsymbol{E}_k\Sigma_{\nu_k}\boldsymbol{E}_k^T, \Sigma_{\xi_k}\}$ and $\Sigma_{\bar{v}_k} = \Sigma_{v_k}$. The initial value $\boldsymbol{x}_0$ satisfies $\boldsymbol{x}_0 \sim \mathcal{N}(\boldsymbol{m}_{x_0}, P_{x_0})$ with $\boldsymbol{m}_{x_0} = [\boldsymbol{m}_{p_0}^T, \boldsymbol{m}_{\omega_0}^T]^T$, $P_{x_0} = \text{diag}\{P_{p_0}, P_{\omega_0}\}$. Without loss of generality, we assume that the noise sequences $\{\boldsymbol{w}_k\}$, $\{\boldsymbol{v}_k\}$, $\{\boldsymbol{\nu}_k\}$, and $\{\boldsymbol{\xi}_k\}$ are mutually independent.

In practice, environmental stiffness $\boldsymbol{\theta}$ is inhomogeneous and varies depending on the position of the contact point, making it unsuitable for offline identification methods. In this work, the environmental stiffness parameter is modeled as time-varying and is identified online in real-time, representing a significant challenge of force estimation.

2) Separability Analysis for the Coupled System: The separation of interaction force disturbances is contingent upon the identification of environmental stiffness $\boldsymbol{\theta}$, necessitating a dataset comprising both the measured force sequence $\boldsymbol{F}_{\text{ext},1:k}$ and the state sequence $\boldsymbol{\omega}_{1:k}$ of the interaction force. A significant challenge arises from the fact that neither the state sequence nor the measurement sequence is directly measurable, thereby emphasizing the importance of their indirect acquisition. The robot–environment coupled system model (14) illustrates the interdependence between interaction forces and robot dynamics, with the nonzero coefficient matrix $\boldsymbol{G}_k$ indicating that the interaction force is as a state-coupled disturbance within the robot dynamics [35]. This interdependence complicates the separability analysis of the interaction force signal by intertwining the estimation of $\boldsymbol{x}_k$ with the identification of $\boldsymbol{\theta}$.

Section III introduces a Bayesian inference-based statistical learning method for identifying stiffness $\boldsymbol{\theta}$, subsequently employing $\hat{\boldsymbol{\theta}}$ to estimate the unmeasurable state $\boldsymbol{\omega}_k$ via a Bayesian filter.

To this end, this issue of separability can be further decomposed into two main aspects: 1) the observability of the augmented state variable $\boldsymbol{x}_k$ from the perspective of control theory; and 2) the identifiability of the stiffness parameter $\boldsymbol{\theta}$ from the perspective of Fisher information theory. Sufficient conditions for the separability of the interaction force $\boldsymbol{F}_{\text{ext},k}$ from the system output can be given as

1) System Observability: The system (14) defined by the pair $(\bar{\boldsymbol{A}}_{\theta_k}, \bar{\boldsymbol{C}}_k)$ is observable if the observability matrix $\mathcal{O}$ has full rank, i.e., $\text{rank}(\mathcal{O}) = \dim(\boldsymbol{x}_k) = n + n_d$. This ensures that the system states are reconstructible from the output, allowing the true value and the prior estimates of the unknown environmental stiffness to be reasonably bounded within specific limits.

2) Force Subsystem Identifiability: The environmental stiffness parameter $\boldsymbol{\theta}$ is locally functionally identifiable if the Fisher Information Matrix (FIM), $\mathcal{I}(\boldsymbol{\theta})$, associated with it has full rank equal to the dimension of $\boldsymbol{\theta}$, i.e., $\text{rank}(\mathcal{I}(\boldsymbol{\theta})) = \dim(\boldsymbol{\theta}) = n_d$ [36]. The FIM is defined using the log-likelihood function $\text{logp}(\boldsymbol{F}_{\text{ext},k})$ and can be calculated as

$$\begin{aligned} \mathcal{I}(\boldsymbol{\theta}) &= \sum_{k=1}^{N} \mathbb{E}\left[\left(\frac{\partial \text{logp}(\boldsymbol{F}_{\text{ext},k};\theta)}{\partial\theta}\right)\left(\frac{\partial \text{logp}(\boldsymbol{F}_{\text{ext},k};\theta)}{\partial\theta}\right)^T\right], \\ &= \sum_{k=1}^{N} \mathbb{E}[\boldsymbol{\omega}_k\boldsymbol{\omega}_k^T] \otimes \Sigma_{vc,k}^{-1}, \end{aligned}$$

where $\otimes$ denotes the Kronecker product. This matrix quantifies the amount of information that the measurements of interaction

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

forces provide about the stiffness parameters, ensuring that $\boldsymbol{\theta}$ is identifiable from the system outputs under the given model assumptions.

III. EEFO for Simultaneous Force Estimation and Stiffness Identification

A. Algorithm Design

This section formulates an online parameter identification problem based on maximum likelihood estimation (MLE). The stiffness estimates are derived by maximizing the log-likelihood function as

$$\hat{\theta} \leftarrow \arg\max_{\theta\in\Theta} \mathcal{L}(\theta) = \arg\max_{\theta\in\Theta} \log p(\boldsymbol{y}_{1:k};\theta) \tag{17}$$

where $\boldsymbol{y}_{1:k} \in \mathbb{R}^{nT}$ denotes the measurements sequence and Θ is the feasible set for parameters. Direct computation of $\mathcal{L}(\theta)$ poses significant challenges. To address this, the EEFO is introduced, employing the EM framework [37] to iteratively estimate the parameter θ and the hidden variables $\boldsymbol{x}_k$. This iterative process facilitates force estimation according to (12).

Consideration is given to the expected complete-data log-likelihood

$$\mathcal{L}(\theta,\theta^*) := \int p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k};\theta^*)\log p(\boldsymbol{x}_k,\boldsymbol{y}_{1:k};\theta)d\boldsymbol{x}_k, \tag{18}$$

where θ^* denotes the current estimate of the stiffness parameter.

Each iteration of the EM algorithm consists of an expectation step (E-step) and a maximization step (M-step). The E-step involves evaluating $\mathcal{L}(\theta,\theta^*)$, primarily by calculating the posterior distribution of $\boldsymbol{x}_k$ using the KF. The log-likelihood function is further detailed as

$$\begin{aligned}&\log p(\boldsymbol{x}_k,\boldsymbol{y}_{1:k};\theta)\\ &= \log p(\boldsymbol{y}_k|\boldsymbol{x}_k) + \log p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta) + \log p(\boldsymbol{y}_{1:k-1}).\end{aligned}$$

Since only the term $\log p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta)$ is dependent on θ, the target function can be redesigned as

$$\mathcal{L}_1(\theta,\theta^*) := \int p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k};\theta^*)\log p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta)d\boldsymbol{x}_k. \tag{19}$$

Consistent with the linear assumption of the force model specified in (12), and given that $p(\boldsymbol{x}_{k-1}|\boldsymbol{y}_{1:k-1};\theta) = \mathcal{N}(\boldsymbol{m}_{k-1}, P_{k-1})$ is Gaussian, and the state transition adheres to a linear Gaussian one, it follows naturally that $p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta)$ also conforms to a Gaussian distribution, expressible as

$$p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta) = \mathcal{N}(\boldsymbol{m}_{k|k-1}, P_{k|k-1}) \tag{20}$$

where $\boldsymbol{m}_{k|k-1} = \bar{\boldsymbol{A}}_\theta \boldsymbol{m}_{k-1} + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}$ and $P_{k|k-1} = \bar{\boldsymbol{A}}_\theta P_{k-1}\bar{\boldsymbol{A}}_\theta^T + \Sigma_{\bar{w},k-1}$. Hence, we have

$$\begin{aligned}&\log p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta)\\ &= c_{P,k|k-1} - \frac{1}{2}(\boldsymbol{x}_k - \boldsymbol{m}_{k|k-1})^T P_{k|k-1}^{-1}(\boldsymbol{x}_k - \boldsymbol{m}_{k|k-1})\\ &= c_{P,k|k-1} - \frac{1}{2}\mathrm{tr}\{P_{k|k-1}^{-1}(\boldsymbol{x}_k\boldsymbol{x}_k^T - \boldsymbol{x}_k\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T\\ &\quad - \boldsymbol{x}_k\boldsymbol{u}_{k-1}^T\bar{\boldsymbol{B}}_{k-1}^T - \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\boldsymbol{x}_k^T - \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}\boldsymbol{x}_k^T\\ &\quad + \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T + \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T\\ &\quad + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1})^T + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T)\}\end{aligned} \tag{21}$$

where $c_{P,k|k-1} = -\frac{1}{2}\log\ |P_{k|k-1}|$.

Taking the expectation of $\log p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1};\theta)$ with respect to $\boldsymbol{x}_k$ allow us to reformulate (19) as follows:

$$\begin{aligned}&\mathcal{L}_1(\theta,\theta^*)\\ &= c_{P,k|k-1} - \frac{1}{2}\mathrm{tr}\{P_{k|k-1}^{-1}(\mathbb{E}[\boldsymbol{x}_k\boldsymbol{x}_k^T] - \mathbb{E}[\boldsymbol{x}_k]\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T\\ &\quad - \mathbb{E}[\boldsymbol{x}_k]\boldsymbol{u}_{k-1}^T\bar{\boldsymbol{B}}_{k-1}^T - \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\mathbb{E}[\boldsymbol{x}_k^T] - \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}\mathbb{E}[\boldsymbol{x}_k^T]\\ &\quad + \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T + \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T\\ &\quad + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1})^T + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T)\}.\end{aligned} \tag{22}$$

In (22), the expectation terms can be expressed as

$$\mathbb{E}[\boldsymbol{x}_k] = \boldsymbol{m}_k, \quad \mathbb{E}[\boldsymbol{x}_k\boldsymbol{x}_k^T] = \boldsymbol{m}_k\boldsymbol{m}_k^T + P_k \tag{23}$$

where $\boldsymbol{m}_k$ and P_k are updated through a KF approach

$$\begin{aligned}\boldsymbol{m}_k^- &= \bar{\boldsymbol{A}}_{\theta^*}\boldsymbol{m}_{k-1} + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}\\ \boldsymbol{m}_k &= \boldsymbol{m}_k^- + K_k(\boldsymbol{y}_k - \bar{\boldsymbol{C}}_k\boldsymbol{m}_k^-)\\ P_k^- &= \bar{\boldsymbol{A}}_{\theta^*}P_{k-1}\bar{\boldsymbol{A}}_{\theta^*}^T + \Sigma_{\bar{w},k-1}\\ K_k &= P_k^-\bar{\boldsymbol{C}}_k^T(\bar{\boldsymbol{C}}_kP_k^-\bar{\boldsymbol{C}}_k^T + \Sigma_{\bar{v}})^{-1}\\ P_k &= (\boldsymbol{I} - K_k\bar{\boldsymbol{C}}_k)P_{k|k-1}P_k^-.\end{aligned} \tag{24}$$

Subsequently, the M-step analytically determines $\hat{\theta}_k$ by applying the gradient descent method to the log-likelihood function $\mathcal{L}_1(\theta,\theta^*)$ with respect to θ.

Define the gradient operator with respect to the stiffness parameter vector

$$\nabla_\theta = \left[\frac{\partial}{\partial\theta_1}, \frac{\partial}{\partial\theta_2}, \ldots, \frac{\partial}{\partial\theta_{nd}}\right]^T. \tag{25}$$

Under standard conditions of continuity and differentiability, the MLE of the stiffness parameters is iteratively updated in the negative gradient direction, according to the following equation:

$$\begin{aligned}\nabla_\theta\mathcal{L}_1(\theta,\theta^*) = \nabla_\theta c_{P,k|k-1} &- \frac{1}{2}\mathrm{trace}\left\{\nabla_\theta P_{k|k-1}^{-1}\cdot\boldsymbol{M}_{\theta,k}\right\}\\ &- \frac{1}{2}\mathrm{trace}\left\{P_{k|k-1}^{-1}\cdot\nabla_\theta\boldsymbol{M}_{\theta,k}\right\}\end{aligned} \tag{26}$$

where $\boldsymbol{M}_{\theta,k}$ is defined as

$$\begin{aligned}\boldsymbol{M}_{\theta,k} := &\mathbb{E}[\boldsymbol{x}_k\boldsymbol{x}_k^T] - \mathbb{E}[\boldsymbol{x}_k]\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T - \mathbb{E}[\boldsymbol{x}_k]\boldsymbol{u}_{k-1}^T\bar{\boldsymbol{B}}_{k-1}^T\\ &- \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\mathbb{E}[\boldsymbol{x}_k^T] - \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}\mathbb{E}[\boldsymbol{x}_k^T]\\ &+ \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T + \bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T\\ &+ \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{A}}_\theta\boldsymbol{m}_{k-1})^T + \bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1}(\bar{\boldsymbol{B}}_{k-1}\boldsymbol{u}_{k-1})^T.\end{aligned} \tag{27}$$

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

Algorithm 1: The EEFO Algorithm.

Initialize $\hat{\theta}_0$, $\boldsymbol{m}_0$, P_0, $\Sigma_{\bar{w}}$, $\Sigma_{\bar{v}}$
for $k = 1, 2, \ldots, T$ **do**
for $j = 1, 2, \ldots, M$ **do**
E-Step:
Update $p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k-1}; \theta)$ according to (20);
Update $p(\boldsymbol{x}_k|\boldsymbol{y}_{1:k}; \theta^*)$ according to (24) given
Update $p(\boldsymbol{x}_{k-1}|\boldsymbol{y}_{1:k-1}; \hat{\theta}_{k-2})$ and $\theta^* = \hat{\theta}_{k-1}$;
Compute $\mathbb{E}[\boldsymbol{x}_k], \mathbb{E}[\boldsymbol{x}_k \boldsymbol{x}_k^T]$ according to (23);
M-Step:
Derive the gradient $\nabla_\theta \mathcal{L}_1(\theta, \theta^*)$ (26);
Find $\hat{\theta}_k$ by solving $\nabla_\theta \mathcal{L}_1(\theta, \theta^*)|_{\theta=\hat{\theta}_k} = 0$;
end for
Return $\hat{\boldsymbol{K}}_{\theta,k} = \text{diag}\{\hat{\theta}_k, \ldots, \hat{\theta}_k\}$;
Return $\hat{\boldsymbol{F}}_{ext,k} = \boldsymbol{E}_k \hat{\boldsymbol{K}}_{\theta,k} \boldsymbol{m}_k$;
end for

The partial derivative terms in (26) can be calculated as

$$\frac{\partial c_{P,k|k-1}}{\partial \theta_i} = -\frac{1}{2}\text{tr}\{P_{k|k-1}^{-1}\frac{\partial P_{k|k-1}}{\partial \theta_i}\}$$

$$\frac{\partial P_{k|k-1}^{-1}}{\partial \theta_i} = -P_{k|k-1}^{-1}\frac{\partial P_{k|k-1}}{\partial \theta_i}P_{k|k-1}^{-1}$$

$$\frac{\partial \boldsymbol{M}_{\theta,k}}{\partial \theta_i} = \frac{\partial \bar{\boldsymbol{A}}_\theta}{\partial \theta_i}\boldsymbol{m}_{k-1}\boldsymbol{m}_{k-1}^T\bar{\boldsymbol{A}}_\theta^T + \bar{\boldsymbol{A}}_\theta \boldsymbol{m}_{k-1}\boldsymbol{m}_{k-1}^T\frac{\partial \bar{\boldsymbol{A}}_\theta^T}{\partial \theta_i} - \mathbb{E}[\boldsymbol{x}_k]\boldsymbol{m}_{k-1}^T\frac{\partial \bar{\boldsymbol{A}}_\theta^T}{\partial \theta_i} - \frac{\partial \bar{\boldsymbol{A}}_\theta}{\partial \theta_i}\boldsymbol{m}_{k-1}\mathbb{E}[\boldsymbol{x}_k^T] \quad (28)$$

where

$$\frac{\partial P_{k|k-1}}{\partial \theta_i} = \frac{\partial \bar{\boldsymbol{A}}_\theta}{\partial \theta_i}P_{k-1}\bar{\boldsymbol{A}}_\theta^T + \bar{\boldsymbol{A}}_\theta P_{k-1}\frac{\partial \bar{\boldsymbol{A}}_\theta^T}{\partial \theta_i}$$

$$\frac{\partial \bar{\boldsymbol{A}}_\theta}{\partial \theta_i} = \begin{bmatrix} \boldsymbol{0} & \boldsymbol{E}_k \boldsymbol{I} \\ \boldsymbol{0} & \boldsymbol{0} \end{bmatrix}, \frac{\partial \bar{\boldsymbol{A}}_\theta^T}{\partial \theta_i} = \left(\frac{\partial \bar{\boldsymbol{A}}_\theta}{\partial \theta_i}\right)^T. \quad (29)$$

For enhanced understanding, the principal procedures of the proposed EEFO algorithm are outlined in Algorithm 1. It is evident that, at each time step, the EEFO operates through repetitive execution of the KF, utilizing the continually updated parameter $\boldsymbol{\theta}$. Consequently, the cumulative computational cost of the proposed EEFO algorithm is quantified as $\mathcal{O}(m^3 \times i)$, where m is the dimension of the augmented state, and i indicates the total number of EM iterations.

B. Simulation Analysis

In this section, the proposed EEFO algorithm undergoes a comprehensive evaluation through Monte Carlo simulations, addressing both constant and varying stiffness scenarios to assess its force estimation performance. The EEFO is benchmarked against five state-of-the-art force estimation methods. Furthermore, the performance of environmental stiffness identification is examined, with a focus on the impact of the number of iteration steps in the EM process of the EEFO.

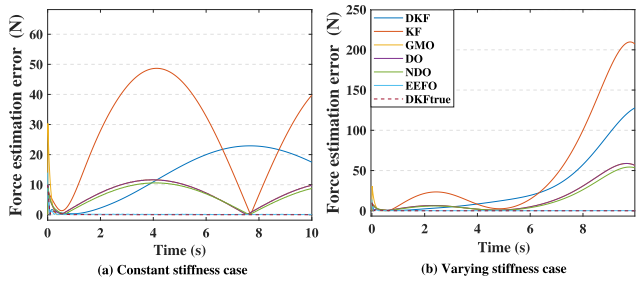


Fig. 4. RMSEs of interaction force estimates from Monte Carlo simulations, for both constant and varying stiffness scenarios.

The interaction force applied to the end-effector of the robot manipulator is generated using a linear stiffness model, with model uncertainties characterized by process and output noises, each following a zero-mean Gaussian distribution with a Standard Deviation (SD) of 0.01N. The sampling interval is set to 0.01s. For the constant scenario, the stiffness parameter is set as $\theta_{\text{con}} = 10$, while for the varying stiffness scenario, the parameter is modeled as a biased sinusoidal signal $\theta(t)_{\text{var}} = 0.5 \cdot \sin(\frac{2\pi t}{T}) + 10$.

1) Evaluation of Force Estimation Performance: A total of 50 Monte Carlo simulations were conducted for both constant and varying stiffness scenarios. The proposed EEFO was compared against five methods: DKF [22], KF [11], GMO [9], NDO [7], and a DO-based force observer [3]. For additional comparison, a DKF with true stiffness was also implemented. All methods were initialized with identical force values. The KF-like methods (KF, DKF, and EEFO) shared the same process and measurement noise variances. The stiffness parameter for DKF was set to $\theta_{\text{DKF}} = 11$, matching the initial stiffness estimate $\hat{\theta}_0$ of the EEFO. Unlike EEFO, DKF employed a time-invariant linear stiffness force model (12) without online stiffness identification or exploration. The gain matrices for NDO, GMO, and DO were configured as $\text{diag}(10, 10, 10)$.

The Root Mean Square Errors (RMSEs) of the interaction force estimation are depicted in Fig. 4. The simulation results indicated that the proposed EEFO and the DKF (with true stiffness) achieve significantly lower estimation errors, attributable to the precise force generation models employed by these methods. The DKF with an incorrect stiffness parameter θ_{DKF} outperforms the KF, suggesting that incorporating environmental characteristics into force estimation is preferable to assuming a constant force signal, provided θ_{DKF} is close to the true value. Specifically, the force generation model utilized by the EEFO, DKF, and NDO proves advantageous in high-dynamic interaction scenarios, which are common in medical-surgical operations.

The distribution of force estimation errors is further illustrated through violin plots in Fig. 5, allowing for a clear comparison of the *precision* and *reliability* of these methods. The mean and median values indicate the precision of force estimation, while the contours of the violin plots depict the error range and density, reflecting the reliability of each method. As shown, the EEFO method demonstrates the highest *precision* and *reliability* in force estimation. In addition, Table I summarizes the Average RMSEs (ARMSEs) and Mean Absolute Errors (MAEs) for

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTIFICATION OF THE FORCE ESTIMATION PERFORMANCE

Scenario	Metric	KF [11]	GMO [9]	DO [3]	NDO [7]	DKF [22]	EEFO (1step)	DKF (true stiffness)
Constant stiffness	ARMSE (N)	16.1950	6.8128	6.6889	6.0375	13.1446	0.1740	0.0967
	MAE (N)	16.1950	6.8107	6.6870	6.0358	13.1446	0.1494	0.0956
Varying stiffness	ARMSE (N)	36.0202	15.1189	14.9942	13.2105	18.3408	0.1776	0.0974
	MAE (N)	36.0202	15.1170	14.9925	13.2089	18.3407	0.1531	0.0964

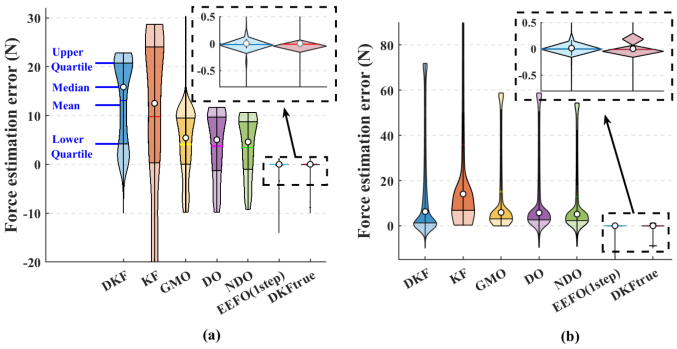


Fig. 5. Violin plots of mean force estimation errors from Monte Carlo simulations. The plot widths at different heights represent the distribution of mean force estimation errors. The darker shaded regions indicate the interquartile range, while the white dots denote the median values for each group. The darkest horizontal lines represent the mean values of the corresponding groups. A magnified view is provided for the proposed EEFO and the DKF with the true stiffness parameter θ.

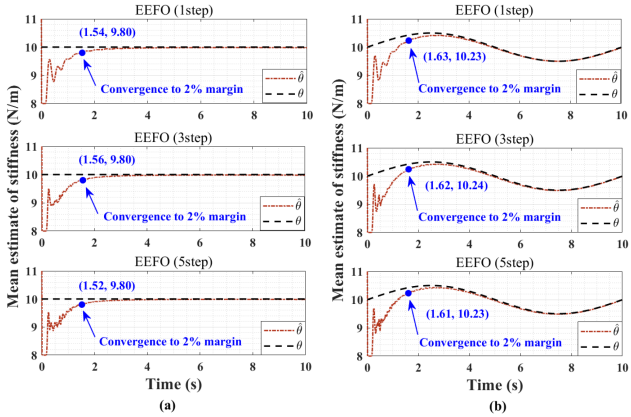


Fig. 6. Mean stiffness estimates from Monte Carlo simulations under constant and varying environmental stiffness scenarios.

force estimation, with the proposed EEFO method achieving the lowest ARMSE and MAE among the six methods, underscoring its superior performance in simultaneous interaction force estimation and stiffness identification.

As a model-based force estimation method, the performance of the proposed EEFO is dependent on the accuracy of the robot dynamics model. Given the inevitable presence of parameter uncertainties, particularly mass uncertainties, in the dynamic model, it is crucial to evaluate and mitigate the parameter sensitivity of the proposed method. To this end, Monte Carlo simulations were conducted with actual parameters varying uniformly within $[-10\%, 10\%]$ of their nominal values. It turns out that the proposed scheme exhibits low sensitivity to perturbations of the mass parameter. The results demonstrate that the proposed scheme exhibits low sensitivity to perturbations in the mass parameter. Moreover, to further reduce the impact of parameter uncertainties, a modified dynamic equation for the robot manipulator can be developed by incorporating a GP-based uncertainty compensation term. Further details on this approach are provided in our previous work [16].

2) Evaluation of Stiffness Identification Performance: The proposed EEFO method was evaluated through 50 Monte Carlo simulations under both constant and varying stiffness scenarios, with EM iteration steps ranging from 1 to 5. The influence of the number of EM iterations on the estimation results was analyzed. As shown in Fig. 6, the fluctuation in the parameter estimates decreases as the number of EM iterations increases for both constant and varying environmental stiffness cases.

In Table II, the stiffness identification results for both constant and varying stiffness scenarios are detailed. In addition to ARMSE, the table includes the convergence time and relative steady-state error to quantitatively assess the transient and steady-state performance of the EEFO algorithm. Convergence time is defined as the time required for the estimated error to first decrease to, and then consistently remain within, a 2% range of its steady-state value. As shown, the parameter estimation errors across all five cases converge within 1.58 s for the constant stiffness scenario and within 1.65 s for the varying stiffness scenario. While the stiffness identification performance varies slightly across the five cases, the ARMSEs and relative steady-state errors for EEFO (1-step) and EEFO (5-step) are lower than for the others. These results underscore the *efficacy* of the proposed scheme in achieving accurate interaction force estimation with online environmental stiffness identification and exploration.

IV. EXPERIMENTAL RESULTS

In this section, the practicality and effectiveness of the proposed scheme are validated through experimental results. In Section IV-A, a calibration experiment conducted on the Universal mechanical testing system (Instron 5543A, USA) is detailed, providing the ground truth of environmental stiffness for the subsequent interaction force estimation experiment. In Section IV-B, experiments are conducted on the Toumai surgical robot platform (MicroPortToumai, CN), where the end effector of a robot interacts with an unknown stiffness deformable environment The proposed EEFO method is compared with five state-of-the-art force estimation methods.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

TABLE II
QUANTIFICATION OF STIFFNESS ESTIMATION PERFORMANCE

		EEFO (1-step)	EEFO (2-step)	EEFO (3-step)	EEFO (4-step)	EEFO (5-step)
Constant stiffness case	ARMSE (N/m)	**0.1464**	0.1607	0.1568	0.1558	0.1559
	Convergence time (s)	1.54	1.58	1.56	1.55	**1.52**
	Relative steady-state error	0.17%	0.25%	0.19%	0.17%	**0.13%**
Varying stiffness case	ARMSE (N/m)	**0.1578**	0.1749	0.1709	0.1699	0.1702
	Convergence time (s)	1.63	1.65	1.62	1.62	**1.61**
	Relative steady-state error	0.17%	0.12%	0.08%	0.08%	**0.07%**

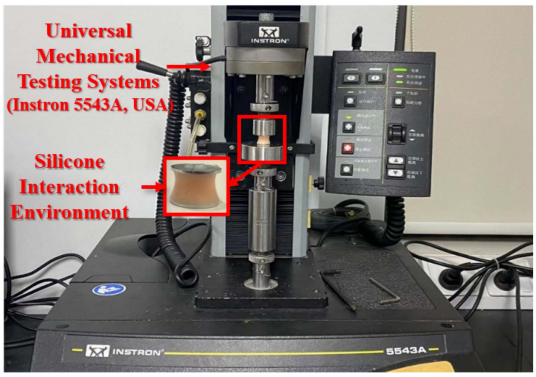


Fig. 7. Experimental setup for environmental stiffness calibration.

A. Environmental Stiffness Calibration Experiments

A linear stiffness model is adopted as the constitutive model of interaction force in Section II-B. The online identification of the unknown stiffness parameter is required, representing one of the significant improvements of the proposed EEFO. To facilitate a better evaluation, a stiffness calibration experiment was carried out in advance to provide the ground truth for the subsequent robot–environment interaction experiments.

In the calibration experiment, a universal mechanical testing system (Instron 5543A, USA) was utilized, as illustrated in Fig. 7. The interaction environment consists of a silicone column, capped on both ends with thin circular steel sheets of 14.77 mm diameter. The ground truth of environmental stiffness was calibrated by compressing the tissue-like silicone column using the universal testing machine. First, the compression force-depth curve was fitted to the model $\boldsymbol{F}_{\text{ext}} = \boldsymbol{K}_\theta \cdot \boldsymbol{\delta}$, resulting in an $R^2 = 0.991$, which demonstrates the high performance of the linear stiffness model in describing the force generation data. From the curve, the stiffness distribution of the silicone column was summarized in a stiffness retrieval table. In the table, the current stiffness can be retrieved according to the deformation degrees of the silicone column. Through this process, the ground truth force generation model was calibrated, which will be used to evaluate the performance of stiffness identification in real time.

B. Interaction Force Estimation Experiments

In this section, the practicality of the proposed EEFO force estimation scheme is validated through experimental tests. The experiment is conducted on a specially designed robot platform, as illustrated in Fig. 8. During the experiment, the end-effector of the slave manipulator, equipped with an ultrasound scalpel, is moved along a predefined interaction trajectory controlled by the drive box. Both static and dynamic interaction scenarios are involved. It should be noted that the residual dynamics were pretrained using a GP with data from offline testing and compensated for in the robot dynamics.

TABLE III
STABILIZATION TIME OF THE FORCE ESTIMATION IN THE EXPERIMENTS

	KF [11]	GMO [9]	DO [3]	NDO [7]	DKF [22]	EEFO
Time (s)	1.65	1.63	1.56	1.56	1.56	**0.57**

The proposed EEFO method is benchmarked against state-of-the-art methods, including KF, GMO, DO, NDO, and DKF. The ground truth is provided by a high-precision force sensor, as shown in Fig. 8. Initial force estimates for all six methods are uniformly set to zero. In addition, model uncertainty and noise level are consistent across the DKF, KF, and EEFO.

As depicted in Fig. 9, the results demonstrate that the proposed EEFO outperforms other methods in terms of transient response, with a faster response time compared to KF and GMO, and reduced fluctuation compared to DO, NDO, and DKF. Furthermore, the stabilization time, specifically during the constant force estimation during (Phase 5 of the interaction process), is summarized in Table III. The analysis reveals that the EEFO reduces stabilization time by over 63.4%.

Fig. 10, illustrates the distribution of force estimation errors through box plots, highlighting that the EEFO method achieves superior *precision* and *reliability*, with estimation errors primarily concentrated around a mean value near zero. In addition, Fig. 11 shows that EEFO delivers lower ARMSE and MAE in interaction force estimation relative to the other methods, aligning with the findings in Section III-B. Notably, the ARMSEs in Fig. 11 indicate that the EEFO scheme enhances force estimation accuracy by 28% compared to DKF, which exhibits the lowest ARMSE among the comparison methods.

To further validate the proposed approach, an additional experiment was conducted where the end-effector was commanded to follow a periodic trajectory during interaction with a silicone column. The results, presented in Fig. 12, demonstrate that all five methods achieve force estimates within 5 s, displaying periodic estimation errors. The EEFO method yielded the smallest steady-state estimation error among the methods compared.

Remark 3: The EEFO method, originally proposed for estimating 3-D forces and demonstrated under the constraints

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

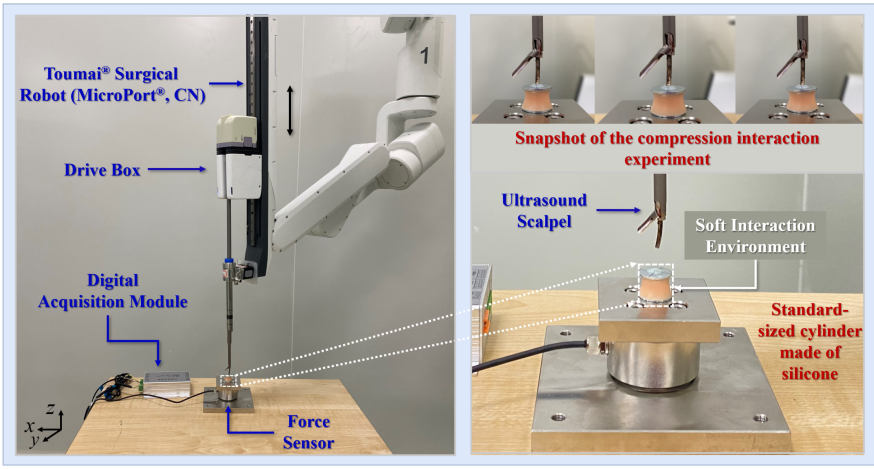


Fig. 8. Experimental setup for interaction force estimation. The research kit comprised a slave manipulator of a Toumai surgical robot (MicroPortToumai, CN) equipped with a common surgical tool, an ultrasound scalpel. A force sensor affixed to the platform provided the ground truth of interaction force. A tissue-like silicone column whose stiffness has been calibrated is placed on the upper surface of the force sensor.

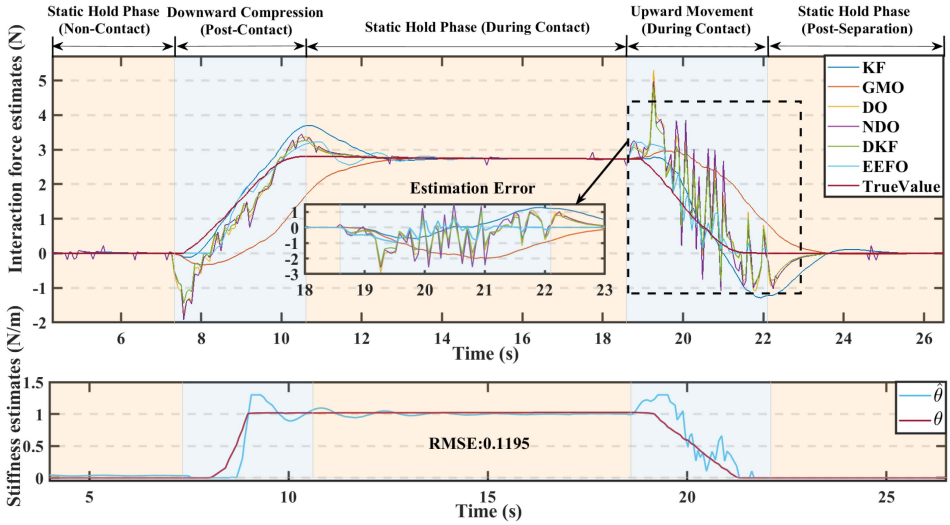


Fig. 9. Experimental results of interaction force estimation and stiffness online estimation. The shaded areas in the figure delineate the five principal stages of the interaction process. 1) Static hold before contact. 2) Constant velocity compression upon initial contact. 3) Static hold during sustained contact. 4) Constant velocity release while in contact. 5) Static hold following contact release.

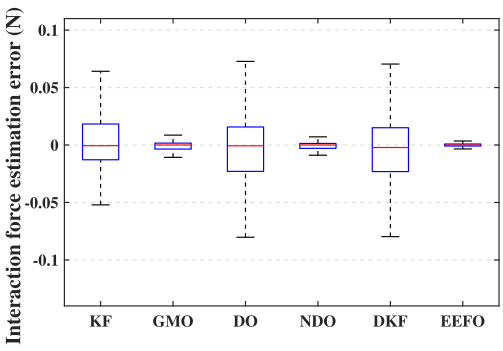


Fig. 10. Box plot of interaction force estimation errors.

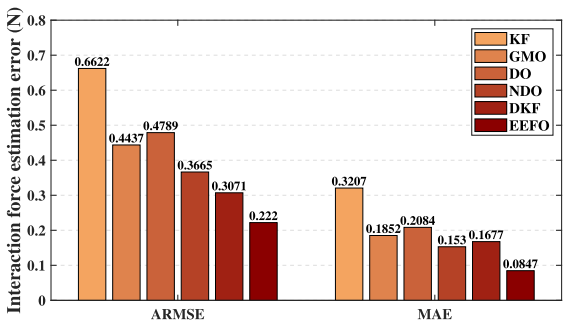


Fig. 11. ARMSEs and MAEs of interaction force estimation experiments.

of a 1-D force sensor, inherently possesses the capability to estimate the complete 6-D Cartesian wrench. Future implementations could extend this approach to full-wrench estimation by integrating a more comprehensive model for environmental force dynamics, thereby accommodating the demands for more complex interaction scenarios.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

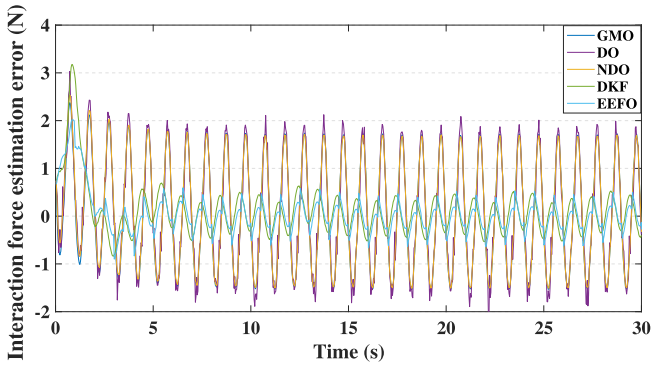


Fig. 12. Experimental results of periodic interaction force estimation.

V. Conclusion

This article investigates the real-time estimation of interaction forces generated by a soft environment with unknown stiffness. To simultaneously exploit information from both robot dynamics and the force generation mechanism, a robot–environment coupled model is first established, and the separability conditions of the interaction force signal are provided. Based on this coupled model, a novel EEFO method is proposed to achieve simultaneous environmental stiffness identification and force estimation iteratively. The proposed framework has been evaluated through numerical simulations in both constant and varying stiffness scenarios and experimentally validated on a surgical robot interacting with tissue-like soft materials. Simulation and experimental results demonstrate that the proposed framework outperforms five state-of-the-art methods, achieving the highest force estimation performance in terms of both *precision* (with an ARMSE improvement of at least 28%) and *reliability* (with the most concentrated estimation errors).

Even though the EEFO method significantly enhances dynamic interaction force estimation capabilities, it still requires further exploration of environmental dynamical properties. This will be addressed in future research, particularly in applications such as beating heart surgery. In addition, applying the EEFO to impedance control in dynamic interaction scenarios [38], and adapting it to low-cost robotic systems, presenting promising avenues for further investigation [39].

References

[1] P. Dupontet al., "A decade retrospective of medical robotics research from 2010 to 2020," *Sci. Robot.*, vol. 6, no. 60, pp. 1–15, 2021.
[2] H. Jiang et al., "A robotic device using gecko-inspired adhesives can grasp and manipulate large objects in microgravity," *Sci. Robot.*, vol. 2, no. 7, pp. 1–12, 2017.
[3] E. Sariyildiz and K. Ohnishi, "An adaptive reaction force observer design," *IEEE/ASME Trans. Mech.*, vol. 20, no. 2, pp. 750–760, Apr. 2015.
[4] S. Katsura, Y. Matsumoto, and K. Ohnishi, "Modeling of force sensing and validation of disturbance observer for force control," *IEEE Trans. Ind. Electron.*, vol. 54, no. 1, pp. 530–538, Feb. 2007.
[5] K. Ohnishi, M. Shibata, and T. Murakami, "Motion control for advanced mechatronics," *IEEE/ASME Trans. Mech.*, vol. 1, no. 1, pp. 56–67, Mar. 1996.
[6] W.-H. Chen, J. Yang, L. Guo, and S. Li, "Disturbance-observer-based control and related methods–An overview," *IEEE Trans. Ind. Electron.*, vol. 63, no. 2, pp. 1083–1095, Feb. 2016.
[7] W.-H. Chen, D. Ballance, P. Gawthrop, and J. O'Reilly, "A nonlinear disturbance observer for robotic manipulators," *IEEE Trans. Ind. Electron.*, vol. 47, no. 4, pp. 932–938, Aug. 2000.
[8] W.-H. Chen, "Disturbance observer based control for nonlinear systems," *IEEE/ASME Trans. Mech.*, vol. 9, no. 4, pp. 706–710, Dec. 2004.
[9] S. Haddadin, A. De Luca, and A. Albu-Schäffer, "Robot collisions: A survey on detection, isolation, and identification," *IEEE Trans. Robot.*, vol. 33, no. 6, pp. 1292–1312, Dec. 2017.
[10] G. Peng, C. Yang, W. He, and C. L. P. Chen, "Force sensorless admittance control with neural learning for robots with actuator saturation," *IEEE Trans. Ind. Electron.*, vol. 67, no. 4, pp. 3138–3148, Apr. 2020.
[11] A. Wahrburg, J. Bös, K. D. Listmann, F. Dai, B. Matthias, and H. Ding, "Motor-current-based estimation of cartesian contact forces and torques for robotic manipulators and its application to force control," *IEEE Trans. Autom. Sci. Eng.*, vol. 15, no. 2, pp. 879–886, Apr. 2018.
[12] T. T. Phuong, K. Ohishi, and Y. Yokokura, "Fine sensorless force control realization based on dither periodic component elimination Kalman filter and wide band disturbance observer," *IEEE Trans. Ind. Electron.*, vol. 67, no. 1, pp. 757–767, Jan. 2020.
[13] L. Roveda, A. A. Shahid, N. Iannacci, and D. Piga, "Sensorless optimal interaction control exploiting environment stiffness estimation," *IEEE Trans. Control Syst. Technol.*, vol. 30, no. 1, pp. 218–233, Jan. 2022.
[14] Y. Wei, W. Li, Y. Yang, X. Yu, and L. Guo, "Decoupling observer for contact force estimation of robot manipulators based on enhanced gaussian process model," in *Proc. IEEE Int. Conf. Cloud Comput. Intell. Syst.*, 2022, pp. 1–7.
[15] R. Cortesão and M. Dominici, "Robot force control on a beating heart," *IEEE/ASME Trans. Mech.*, vol. 22, no. 4, pp. 1736–1743, Aug. 2017.
[16] Y. Wei, S. Lyu, W. Li, X. Yu, Z. Wang, and L. Guo, "Contact force estimation of robot manipulators with imperfect dynamic model: On Gaussian process adaptive disturbance Kalman filter," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3524–3537, Jul. 2024.
[17] Y. Han, J. Wu, C. Liu, and Z. Xiong, "An iterative approach for accurate dynamic model identification of industrial robots," *IEEE Trans. Robot.*, vol. 36, no. 5, pp. 1577–1594, Oct. 2020.
[18] D. Nguyen-Tuong and J. Peters, "Model learning for robot control: A survey," *Cogn. Process.*, vol. 12, no. 4, pp. 319–340, 2011.
[19] Y. Zhu, J. Qiao, and L. Guo, "Adaptive sliding mode disturbance observer-based composite control with prescribed performance of space manipulators for target capturing," *IEEE Trans. Ind. Electron.*, vol. 66, no. 3, pp. 1973–1983, Mar. 2019.
[20] L. Guo and S. Cao, *Anti-Disturbance Control for Systems With Multiple Disturbances*. New York, NY, USA: CRC Press, 2013.
[21] C. B. Yigit, E. Bayraktar, O. Kaya, and P. Boyraz, "External force/torque estimation with only position sensors for antagonistic VSAs," *IEEE Trans. Robot.*, vol. 37, no. 2, pp. 675–682, Apr. 2021.
[22] J. Hu and R. Xiong, "Contact force estimation for robot manipulator using semiparametric model and disturbance Kalman filter," *IEEE Trans. Ind. Electron.*, vol. 65, no. 4, pp. 3365–3375, Apr. 2018.
[23] L. Guo, W. Li, Y. Zhu, X. Yu, and Z. Wang, "Composite disturbance filtering: A novel state estimation scheme for systems with multi-source, heterogeneous, and isomeric disturbances," *IEEE Open J. Ind. Electron. Soc.*, vol. 4, pp. 387–400, 2023.
[24] L. Santos and R. Cortesão, "Computed-torque control for robotic-assisted tele-echography based on perceived stiffness estimation," *IEEE Trans. Autom. Sci. Eng.*, vol. 15, no. 3, pp. 1337–1354, Jul. 2018.
[25] N. Diolaiti, C. Melchiorri, and S. Stramigioli, "Contact impedance estimation for robotic systems," *IEEE Trans. Robot.*, vol. 21, no. 5, pp. 925–935, Oct. 2005.
[26] R. Schindeler and K. Hashtrudi-Zaad, "Online identification of environment Hunt–Crossley models using polynomial linearization," *IEEE Trans. Robot.*, vol. 34, no. 2, pp. 447–458, Apr. 2018.
[27] L. Wijayarathne, Z. Zhou, Y. Zhao, and F. L. Hammond, "Real-time deformable-contact-aware model predictive control for force-modulated manipulation," *IEEE Trans. Robot.*, vol. 39, no. 5, pp. 3549–3566, Oct. 2023.
[28] Y. Zhang, Y. Yang, Y. A. Tse, and M. Y. Wang, "Effective estimation of contact force and torque for vision-based tactile sensors with Helmholtz–Hodge decomposition," *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 4094–4101, Oct. 2019.
[29] L. Roveda and D. Piga, "Sensorless environment stiffness and interaction force estimation for impedance control tuning in robotized interaction tasks," *Auton. Robot.*, vol. 45, no. 3, pp. 371–388, 2021.
[30] S. Särkkä, Ed., *Bayesian Filtering and Smoothing*. Otakaari, Finland: Cambridge Univ. Press, 2013.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.

[31] M. Su et al., "A new micro-traumatic Laparoscopic surgery robot system," *Chin. J. Med. Instrum.*, vol. 43, no. 3, pp. 165–169, 2019.
[32] B. Siciliano and O. Khatib, Eds., *Springer Handbook of Robotics*. Amsterdam, Netherlands: Springer, 2008.
[33] G. Gilardi and I. Sharf, "Literature survey of contact dynamics modelling," *Mech. Mach. Theory*, vol. 37, no. 10, pp. 1213–1239, 2002.
[34] P. Moreira, N. Zemiti, C. Liu, and P. Poignet, "Viscoelastic model based force control for soft tissue interaction and its application in physiological motion compensation," *Comput. Methods Programs Biomed.*, vol. 116, no. 2, pp. 52–67, 2014.
[35] C. Wang, L. Guo, C. Wen, Y. Zhu, and J. Qiao, "Adaptive anti-disturbance control for a class of uncertain nonlinear systems with composite disturbances," *IEEE Trans. Cybern.*, vol. 54, no. 7, pp. 4241–4254, Jul. 2024.
[36] T. J. Rothenberg, "Identification in parametric models," *Econometrica*, vol. 39, no. 3, pp. 577–591, 1971.
[37] W. Li, B. Tian, J. Qiao, and L. Guo, "Estimating quasiperiodic disturbance with unknown frequency via expectation–maximization," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 6843–6856, Jul. 2022.
[38] L. Roveda, F. Vicentini, and L. M. Tosatti, "Deformation-tracking impedance control in interaction with uncertain environments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2013, pp. 1992–1997.
[39] C. B. Yigit, E. Bayraktar, and P. Boyraz, "Low-cost variable stiffness joint design using translational variable radius pulleys," *Mech. Mach. Theory*, vol. 130, pp. 203–219, 2018.



Yanran Wei received the B.Eng. degree in process equipment and control engineering from Dalian University of Technology, Dalian, China, in 2018. She is currently working toward the Ph.D. degree in control theory and control engineering from the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China.

Her research interests include antidisturbance control and filtering, Bayesian estimation, haptic force estimation, and robot learning.



Jiayin Wang received the B.S. and M.S. degrees in mechanical design, manufacturing, and automation from Shandong University, Jinan, China, in 2011 and 2014, respectively. He is currently working toward the Ph.D. degree in electronic information engineering with the School of Electronic Information Engineering, Tongji University, Shanghai, China.

He is currently the Senior R&D Director with the Shanghai MicroPort MedBot (Group) Company Ltd. His research interests include surgical robotics, teleoperation control systems, force control, and robotic kinematics and dynamics.

Mr. Wang was the recipient of the First Prize of the Shanghai Science and Technology Progress Award.



Wenshuo Li received the B.Eng. degree in automation from Shandong University, Jinan, China, in 2012, and the Ph.D. degree in control theory and control engineering from Beihang University, Beijing, China, in 2019.

From 2016 to 2017, he was a Joint Ph.D. Student with the Department of Computer Science, Brunel University London, London, U.K. He is currently a Research Fellow with Hangzhou Innovation Institute, Beihang University, Hangzhou, China. His research interests include nonlinear filtering, antidisturbance control and filtering, Bayesian estimation, and composite disturbance filtering with applications in autonomous navigation systems.



Xiaofan Du received the B.S. degree in aircraft control and information engineering from the Northwestern Polytechnical University, Xi'an, China, in 2023. She is currently working toward the M.S. degree in control theory and control engineering with the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China.

Her current research interests include optimal filtering, antidisturbance estimation and their applications in intelligent navigation systems.



Xiang Yu (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in automation science and engineering from Northwestern Polytechnical University, Xi'an, China, in 2003, 2004, and 2008, respectively.

He is currently a Professor with the School of Automation Science and Electrical Engineering, Beihang University, Beijing, China. He was a Postdoctoral Research Fellow with the Department of Electrical and Computer Engineering, Western University, London, ON, Canada, and a Research Associate with the Department of Mechanical, Industrial and Aerospace Engineering, Concordia University, Montreal, QC, Canada. He has authored more than 80 prestigious journal papers and a monograph, including IEEE transactions, AIAA journals, and *Progress in Aerospace Sciences*. His current research interests include safety control of unmanned aerial vehicles, autonomous navigation and control of flight vehicles.

Prof. Yu was the recipient of the Recruitment Program for Young Professionals, First Prize of Science and Technology Progress Award of China Instrument and Control Society, Youth Science and Technology Award of Chinese Society of Aeronautics and Astronautics, Gold Medal for outstanding performances at the 48th International Exhibition of Inventions of Geneva 2023, and the Best Paper and Best Paper Finalist at international conferences. He has also served as the Associate Editors of four journals.



Lei Guo (Fellow, IEEE) was born in Qufu, China, in 1966. He received the B.S. degree in mathematics and M.S. degree in operations research and cybernetics from Qufu Normal University, Qufu, in 1988 and 1991, respectively, and the Ph.D. degree in automatic control and applications from Southeast University, Nanjing, China, in 1997.

From 1991 to 1994, he was a Lecturer with Qingdao University, Qingdao, China. He has published more than 480 papers, seven monographs, and has more than 180 authorized invention patents. His research interests include anti-disturbance control theory and applications, intelligent navigation and control technology of unmanned systems.

Prof. Guo was a Postdoctoral Fellow and Associate Fellow with Southeast University, from 1997 to 1999. From 1999 to 2000, he was a Postdoctoral Fellow with IRCCyN, Nantes, Frances. From 2000 to 2005, he was a Research Associate/Fellow/Visiting Professor with Loughborough University, UMIST and Manchester University, U.K. From 2004 to 2006, he was a Professor with Southeast University, now he is a Guest Chair Professor. From 2006 to now, he is a Distinguish Professor and Director of the Space Intelligent Control Research Center, Beihang University, Beijing, China. He is an Academician of the Chinese Academy of Sciences (CAS), a Fellow of IEEE, IET, Chinese Association of Automation (CAA), and China Association of Inventions (CAI). He is the Director of the Navigation, Guidance and Control Committee of the CAA. He was the recipient of the National Nature Science Awards (2013), National Technology Invention Awards (2018), National Pioneer Innovation Award (2023) of China. He also obtained the Gold Medal of International Exhibition of Inventions of Geneva, Nuremberg and Turkey for bio-inspired navigation sensor, compound-eye-inspired navigation systems and biomimetic flying robots, respectively.

Authorized licensed use limited to: UNIVERSITY OF SOUTHAMPTON. Downloaded on October 31,2026 at 17:21:55 UTC from IEEE Xplore. Restrictions apply.